

261100690008
Prathiksha Patil

command line arguments, fileoperators, read and filters

Write a Bash script that accepts a filename as the first command-line argument.
Check whether an argument is provided.
Check whether the file exists using a file operator.
If the file exists, display the first 5 lines using head.
Otherwise, display an appropriate error message.
File Type and Permission Checker

```
#!/bin/bash
```

```
if [ $# -eq 0 ]  
then  
    echo "Please provide a filename."  
elif [ -f "$1" ]  
then  
    echo "First 5 lines of the file:"  
    head -n 5 "$1"  
else  
    echo "File does not exist."  
fi
```

Write a Bash script that accepts a path as a command-line argument and determines whether it is:
a regular file,
a directory,
readable,
writable, and
executable.
Display suitable messages using file operators such as -f, -d, -r, -w, and -x.

```
#!/bin/bash
```

```
if [ $# -eq 0 ]  
then  
    echo "Please provide a path."  
    exit  
fi
```

```
if [ -f "$1" ]  
then  
    echo "It is a regular file."  
fi
```

```
if [ -d "$1" ]  
then
```

```

    echo "It is a directory."
fi

if [ -r "$1" ]
then
    echo "It is readable."
else
    echo "It is not readable."
fi

if [ -w "$1" ]
then
    echo "It is writable."
else
    echo "It is not writable."
fi

if [ -x "$1" ]
then
    echo "It is executable."
else
    echo "It is not executable."
fi

```

Write a Bash script that accepts a log file as a command-line argument and searches for the words:

failed, denied, unauthorized, and error.

```
#!/bin/bash
```

```

if [ $# -eq 0 ]
then
    echo "Please provide a log file."
    exit
fi

if [ ! -f "$1" ]
then
    echo "Log file does not exist."
    exit
fi

grep -iE "failed|denied|unauthorized|error" "$1"

```

Write a Bash script that accepts a filename and checks whether:
the file exists,
it is a regular file,
it is empty or non-empty.

If it is non-empty, display the first 5 lines.

```
#!/bin/bash
```

```
if [ $# -eq 0 ]  
then  
    echo "Please provide a filename."  
    exit  
fi
```

```
if [ ! -e "$1" ]  
then  
    echo "File does not exist."  
    exit  
fi
```

```
echo "File exists."
```

```
if [ -f "$1" ]  
then  
    echo "It is a regular file."  
else  
    echo "It is not a regular file."  
    exit  
fi
```

```
if [ -s "$1" ]  
then  
    echo "File is non-empty."  
    echo "First 5 lines:"  
    head -n 5 "$1"  
else  
    echo "File is empty."  
fi
```

Write a Bash script named investigate.sh that accepts a log file as a command-line argument.

The script must first verify that the file exists and is readable. Then ask the investigator to enter a search term.

Display the first 5 matching records and report the total number of matching records.

```
#!/bin/bash
```

```
if [ $# -eq 0 ]  
then  
    echo "Please provide a log file."  
    exit  
fi
```

```
if [ ! -e "$1" ]
```

```
then
    echo "Log file does not exist."
    exit
fi

if [ ! -r "$1" ]
then
    echo "Log file is not readable."
    exit
fi

read -p "Enter search term: " term

echo "First 5 matching records:"
grep -i "$term" "$1" | head -n 5

count=$(grep -ic "$term" "$1")

echo "Total matching records: $count"
```